RAM POLINENI

NuStudio.ai

AI/ML & GenAI Architect | Agentic AI | Multi-Agent Systems
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SUMMARY

Strategic AI/ML, NLP, GenAI and software engineering professional with around 15 years of experience delivering enterprise applications, predictive models and cloud-based AI solutions. Strong career progression from full-stack development and middleware integration to machine learning, NLP, Retrieval-Augmented Generation and Agentic AI architecture. Experienced in designing production-grade multi-agent systems, enterprise knowledge assistants, intelligent document-processing platforms and predictive-maintenance solutions. Proven ability to lead cross-functional teams, define solution architecture and translate complex business requirements into practical technical solutions. Strong background in technical leadership, stakeholder engagement, production delivery and business-focused AI transformation.

TECHNICAL SKILLS

GenAI and Agentic AI: LLMs, Agentic AI, Multi-Agent Systems, End-to-End RAG Pipelines, LangGraph, LangChain, Prompt Engineering, Context Engineering, Tool Calling, Structured Outputs, Workflow Orchestration, Agent Memory, Human-in-the-Loop, MCP, Embeddings, Vector Search, Hybrid Search, Semantic Reranking, Fine-Tuning, LLM and RAG Evaluation, Guardrails, Observability, Audit Logging

AI/ML/NLP: Machine Learning, Deep Learning, Regression, Classification, Clustering, Time-Series Forecasting, Anomaly Detection, Predictive Maintenance, BERT, Named Entity Recognition, Text Classification, Information Extraction, Text Summarization, Feature Engineering, Model Explainability

Programming and Application Engineering: Python, Java 17, SQL, FastAPI, Spring Boot, Spring Data JPA, Spring Security, Hibernate, REST APIs, SOAP Web Services, Microservices Architecture, Angular and API Integration

Azure: Azure OpenAI, Azure AI Search, Azure AI Document Intelligence, Azure Kubernetes Service, Azure Functions, Azure Service Bus, Azure API Management, Microsoft Entra ID

AWS: Amazon SageMaker, Amazon Textract, AWS Lambda, AWS Step Functions, Amazon Kinesis, AWS Glue, Amazon S3

GCP: Vertex AI, Vertex AI Search, Vertex AI Vector Search, Document AI, Google Kubernetes Engine, Cloud Run, Apigee

ML Frameworks: Scikit-learn, XGBoost, LightGBM, Isolation Forest, TensorFlow, PyTorch, Hugging Face Transformers, spaCy, Pandas, NumPy and MLflow

Data and Enterprise Integration: PostgreSQL, Oracle, Redis, Apache Kafka, SharePoint Integration, ETL Pipelines, Data Processing, Event-Driven Architecture, Asynchronous Processing and Middleware Integration

MLOps/LLMOps: MLflow, Model Registry, Model and Prompt Versioning, Automated Evaluation, CI/CD, Docker, Kubernetes, OpenShift, Azure DevOps, GitHub Actions, Model Monitoring, Data Drift, Model Drift, Production Observability

EXPERIENCE

Principal Architect – Data & AI Aug 2026 - Present
Nustudio.ai (Full-time)

Solutions Architect Nov 2011 – Aug 2026
Cognizant Technology Solutions India Pvt Ltd (Full-time)

PROJECTS

Project 1: Azure-Based Agentic AI Platform for IT Incident Resolution

Role: Principal GenAI Architect | Domain: Automobile E-Commerce

Developed an Azure-based Agentic AI platform to automate the investigation and resolution of recurring production incidents across enterprise applications, databases and cloud infrastructure. The solution analysed monitoring alerts, application logs, configuration data, recent deployments and historical incidents to identify probable root causes and execute approved remediation workflows.

- Designed the end-to-end architecture using Azure OpenAI, LangGraph-based agent orchestration, Azure API Management, Azure Service Bus, Azure Functions and Azure Kubernetes Service.

- Developed specialized incident-triage, application-diagnostic, infrastructure-diagnostic, root-cause-analysis and remediation agents with shared workflow state, conditional routing, retries and escalation paths.
- Implemented automated investigation workflows for application outages, failed batch jobs, API failures, database connectivity issues, resource-utilization alerts, certificate expiry and Kubernetes workload failures.
- Integrated the agents with ServiceNow, Azure Monitor, Log Analytics and enterprise monitoring APIs to collect incident details, configuration information, logs, metrics, dependency status and recent deployment history.
- Developed remediation workflows using Azure Functions and Azure Automation to perform approved actions such as restarting services, scaling workloads, retrying failed jobs, rotating credentials and rolling back application deployments.
- Implemented input, output and tool-execution guardrails to detect prompt injection, validate agent recommendations and restrict agents to approved APIs, parameters, environments and remediation actions.
- Introduced confidence thresholds and risk-based human approval, allowing autonomous execution only for predefined low-risk incidents while maintaining complete audit records of agent decisions and actions.
- Deployed the platform on Azure Kubernetes Service with Microsoft Entra ID authentication, Azure Key Vault secrets management, Azure Cache for Redis workflow state, Application Insights monitoring and Azure DevOps CI/CD.

Technology Stack: Python, FastAPI, LangGraph, Azure OpenAI, Azure Kubernetes Service, Azure API Management, Azure Service Bus, Azure Functions, Azure Automation, Azure Monitor, Log Analytics, Azure Cache for Redis, Microsoft Entra ID, Azure Key Vault, Azure DevOps and ServiceNow APIs.

Project 2: Enterprise RAG Knowledge Assistant

Role: AI/ML Architect | Domain: Healthcare

Developed an enterprise Retrieval-Augmented Generation platform entirely on Azure to provide employees with grounded answers from approved policies, procedures, technical documents and operational manuals. The platform returned document-level and page-level citations while enforcing access restrictions based on user role and department.

- Designed the end-to-end Azure RAG architecture covering document ingestion, parsing, chunking, embedding generation, indexing, retrieval, reranking, response generation, citations and evaluation.
- Developed ingestion workflows for SharePoint, Azure Blob Storage, internal applications and structured databases, including incremental updates, duplicate detection and removal of superseded document versions.
- Implemented document parsing and OCR using Azure AI Document Intelligence for PDFs, Word documents, presentations, spreadsheets and scanned files while preserving headings, sections, page numbers and metadata.
- Developed structure-aware and semantic chunking strategies and indexed document chunks, embeddings, source references and access metadata in Azure AI Search.
- Implemented hybrid retrieval using vector search, keyword search, metadata filtering, query rewriting and semantic reranking to improve results for exact policy terms and natural-language questions.
- Developed grounded prompt templates using Azure OpenAI that required responses to be generated only from retrieved evidence and return validated citations or a controlled fallback when sufficient information was unavailable.
- Implemented RAG guardrails covering indirect prompt-injection detection, retrieval relevance thresholds, citation validation, PII masking and document-level authorization before content was passed to the language model.
- Developed FastAPI-based ingestion and chat services and deployed them using Azure Container Apps with Microsoft Entra ID, Azure Key Vault, Azure Cache for Redis, Application Insights and Azure DevOps pipelines.

Technology Stack: Python, FastAPI, LangChain, Azure OpenAI, Azure AI Search, Azure AI Document Intelligence, Azure Blob Storage, SharePoint, Azure Container Apps, Azure Cache for Redis, Microsoft Entra ID, Azure Key Vault, Application Insights and Azure DevOps.

Project 3: AWS-Based Predictive Maintenance and Anomaly-Detection Platform

Role: Lead Data Scientist | Domain: Manufacturing and Industrial Operations

Developed an AWS-based predictive-maintenance platform that analysed equipment sensor data, operating conditions and maintenance history to identify abnormal machine behaviour and predict potential failures before production was affected.

- Designed streaming and batch data pipelines using Amazon Kinesis, AWS Glue and Amazon S3 to process temperature, vibration, pressure, current, voltage, equipment load and operational-status data.
- Implemented data-quality and preprocessing workflows to handle missing readings, duplicate events, invalid sensor values, timestamp misalignment and changes in machine operating conditions.
- Developed rolling-window, lag, rate-of-change, frequency-domain, utilization and maintenance-history features to represent equipment degradation and failure patterns.

- Built failure-prediction models using XGBoost and Isolation Forest and developed anomaly-detection models using Isolation Forest and autoencoders.
- Addressed imbalanced failure data using class weighting, training-data resampling and threshold optimization while applying time-based validation to prevent data leakage.
- Evaluated model performance using precision, recall, F1-score, PR-AUC, false-alert rate and failure-warning lead time, with separate thresholds for critical and non-critical assets.
- Implemented prediction guardrails using confidence thresholds, SHAP-based explainability and human approval to prevent uncertain model outputs from directly triggering critical maintenance activities.
- Used Amazon SageMaker for model training, experiment tracking, model registration and managed inference endpoints, with AWS Lambda, API Gateway and Amazon CloudWatch supporting integration and monitoring.

Technology Stack: Python, Scikit-learn, XGBoost, Isolation Forest, TensorFlow, Amazon Kinesis, AWS Glue, Amazon S3, Amazon SageMaker, AWS Lambda, Amazon API Gateway and Amazon CloudWatch.

Project 4: AI Document Intelligence Platform

Role: Senior Data Scientist | Domain: Logistics and Supply Chain

Developed a GCP-based document intelligence platform to automate the processing of shipment documents received from suppliers, carriers and freight forwarders, including commercial invoices, packing lists, bills of lading, certificates of origin and customs declarations. The solution classified and extracted shipment information, reconciled details across related documents and routed validated data and discrepancies to logistics and customs-compliance teams.

- Designed an event-driven processing pipeline using Cloud Storage, Eventarc, Pub/Sub, Cloud Run and Workflows to support scalable asynchronous processing of shipment document sets received through email and supplier portals.
- Implemented OCR, table extraction and layout parsing using Google Document AI, including splitter processors to separate combined shipment files into individual documents and extract line items from invoices and packing lists.
- Developed document-classification models using TF-IDF and gradient boosting, and later fine-tuned BERT-based models using Vertex AI custom training for complex and visually similar document categories.
- Developed named entity recognition and information-extraction models to identify shipper and consignee details, container numbers, bill of lading references, HS codes, port names, item descriptions, quantities, weights, declared values, Incoterms and dates.
- Built cross-document reconciliation rules to compare quantities, weights and declared values across commercial invoices, packing lists and bills of lading, flagging mismatches before customs filing.
- Implemented validation logic using field-level confidence thresholds, business-rule validation and an operations review queue before verified information was transferred to transport management and customs filing systems.
- Captured reviewer corrections and processing outcomes in BigQuery to support model evaluation, error analysis and scheduled retraining through Vertex AI Pipelines and Model Registry. Exposed services through Apigee and monitored volume, latency, failures and model confidence using Cloud Monitoring and Cloud Logging.

Technology Stack: Python, Scikit-learn, spaCy, Google Document AI, Vertex AI, Vertex AI Pipelines, Cloud Storage, Pub/Sub, Cloud Run, Workflows, BigQuery, Apigee, Cloud Monitoring and Cloud Logging.

Project 5: Enterprise Full-Stack Application and Middleware Integration Platform

Role: Senior Full-Stack Java Developer | Domain: Enterprise Transaction and Workflow Management

Developed an end-to-end enterprise application to manage customer requests, business transactions, approval workflows, document processing and operational reporting. The platform integrated with CRM applications, payment services, document repositories and legacy systems through synchronous and event-driven middleware services.

- Designed the application architecture covering Angular user interfaces, Spring Boot microservices, secured REST APIs, middleware integrations, relational databases and container-based deployment.
- Developed Spring Boot services for customer management, transaction processing, approval workflows, document handling and operational reporting using Spring Data JPA and Hibernate.
- Implemented REST and SOAP integrations to connect the application with CRM platforms, payment services, document-management systems and legacy applications.
- Developed Kafka-based middleware services for message validation, transformation, enrichment, orchestration and routing between enterprise systems.
- Implemented resilience patterns including timeouts, retries, circuit breakers, fallback processing, centralized exception handling and idempotent transaction execution.

- Implemented Spring Security, OAuth 2.0 and JWT-based authentication with role-based access to application functions, APIs and business data.
- Developed Angular modules for transaction processing, approval workflows, search and reporting and integrated them with secured backend services.
- Containerized the application and deployed it on OpenShift with GitHub Actions-based CI/CD, automated testing, SonarQube quality checks, centralized ELK logging and Spring Boot Actuator monitoring.

Technology Stack: Java 17, Spring Boot, Spring Data JPA, Spring Security, Spring Cloud, Hibernate, REST APIs, SOAP Web Services, Angular, Apache Kafka, Redis, Oracle, Maven, JUnit, Swagger/OpenAPI, GitHub Actions, Docker, OpenShift and ELK Stack.

EDUCATION

BE (Bachelor of Engineering)
Tagore Engineering College, Chennai